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SciRerankBench: Benchmarking Rerankers Towards Scientific Retrieval-Augmented Generated LLMs

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Abstract

Scientific literature question answering is a pivotal step towards new scientific discoveries. Recently, *two-stage* retrieval-augmented generated large language models (RAG-LLMs) have shown impressive advancements in this domain. Such two-stage framework, especially the second stage (*reranker*), is particularly essential towards scientific domain, where subtle differences in terminology may have greatly negative impacts on the final factual-oriented or knowledge-intensive answers. Despite this significant progress, the potential and limitations of these works remain unexplored. In this work, we present a *Sci*entific *Rerank*-oriented RAG *Bench*mark (***SciRerankBench***), for evaluating rerankers within RAG-LLMs systems, spanning 5 scientific subjects.It is derived from over 250 million scholarly works with more than 100 million authors. To rigorously assess the reranker performance in terms of noise resilience, relevance disambiguation, and factual consistency, we develop 3 types of question-context-answer *(Q-C-A)* pairs, i.e., *Noisy Contexts (NC)*, *Semantically Similar but Logically Irrelevant Contexts (SSLI)*, and *Counterfactual Contexts (CC)*. Through systematic evaluation of 13 widely used rerankers on 5 families of LLMs, we provide detailed insights into their relative strengths and limitations. To the best of our knowledge, SciRerankBench is the first benchmark specifically developed to evaluate rerankers within RAG-LLMs, which provides valuable observations and guidance for their future development.

CCS Concepts

• **Computing methodologies** → **Search methodologies**; *Natural language generation.*

Keywords

RAG, LLM, Reranker, Reranking, Scientific Literature, Retrieval Augmented Generation

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1 Introduction

Scientific literature question answering has always been a pivotal steps for scientific new discoveries [12, 14, 16, 27, 28, 62]. For example, a biologist studying CRISPR gene-editing must consult hundreds of papers to understand off-target effects [3, 15]. In COVID-19 research [46, 57], clinicians have to rapidly analyze thousands of emerging papers each week to make informed public health decisions. Recently, retrieval-augmented generated large language models (RAG-LLMs) [1, 26, 59] have gained substantial progress in scientific literature analysis. By effectively combining retrieval and generative capabilities, RAG-LLMs demonstrate significant advantages when addressing knowledge-intensive, factually-oriented, and reliability-critical tasks [9, 18, 51]. This integration allows models to improve factual accuracy while generating responses that are more interpretable and reliable, making them well-suited for scientific tasks requiring trustworthy and rigorous reasoning.

Current RAG-LLMs generally adopts a *two-stage* pipeline. The first stage conducts fast but low-precision (*embedding model*) retrievals, while the second stage (*rerankers*) emphasizes high-precision but higher computational costs for a trade-off between effectiveness and efficiency. Such two-stage framework, especially the second stage, is particularly essential in the scientific domain, where subtle differences in terminology may have greatly negative impacts on the final factual-oriented or knowledge-intensive answers [19, 24, 30, 36]. As illustrated in Figure 1, sorts of widely-used rerankers, such as BCE [31], BGE [6], MXBAI [38] and ColBert [37], have been developed into mature products and have received significant attention from both academia and industry.

Despite significant attention, the capabilities of rerankers in scientific domains remain largely unexplored. Recent research primarily investigates the overall RAG-LLM systems, focusing predominantly on the quality of their final outputs, and neglects evaluating the effectiveness of individual components. For example, Rankify [1] and FlashRAG [18] provide a comprehensive and modular Python toolkit for retrieval, reranking, and generation, supporting reproducible experimentation. Johannesson and Olsson [22] propose a benchmark to evaluate re-ranking methods in LLM-based search systems. However, these studies notably overlook detailed evaluation of the reranking module. Furthermore, their datasets are sourced from general domains, thus failing to effectively assess the deeper reasoning capabilities required by RAG-LLMs in the scientific domain. Other researchers concentrate on constructing

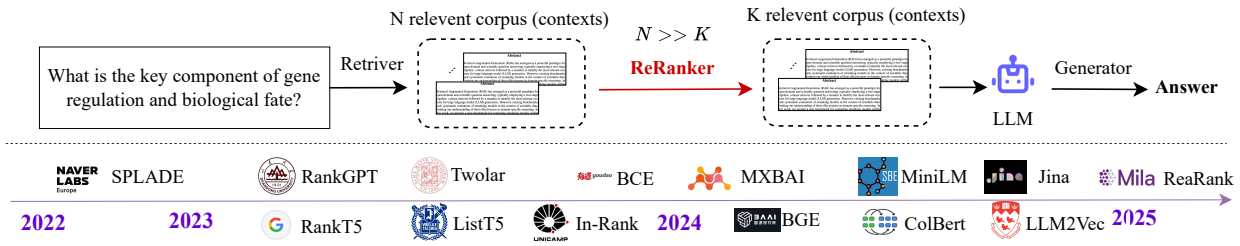


Figure 1: Overall architecture of the *two-stage* RAG-LLM pipeline. The first stage conducts fast but low-precision (*embedding model*) retrievals, while the second stage (*rerankers*) emphasizes high-precision but higher computational costs for a trade-off between effectiveness and efficiency in RAG-LLMs.This two-stage setup first narrows the candidate set efficiently, then rerank results for higher quality generation.

scientific question answering (QA) benchmarks without incorporating retrieval-augmented generation, meaning their evaluations do not involve external knowledge as input contexts for LLMs. Typical works include SciEval [40], SciBench [48], AGIEval [61], SciHorizon [35], MATH [13], and AquaRat [25] all of which aim to systematically evaluate AI-for-Science models and datasets. However, these benchmarks neglect external retrieved contexts, making them unsuitable for evaluating the deep reasoning capabilities of RAG-LLMs when addressing knowledge-intensive, fact-oriented, and high-reliability tasks. Table 1 provides a detailed comparison between this paper and related studies.

In response, we propose a *Sci*entific *Rerank*-oriented RAG *Bench*mark (***SciRerankBench***). It is derived from over 250 million scholarly works with more than 100 million authors [33]. Our synthetic dataset contains 4.5K question-context-answer (Q-C-A) pairs, which spanning 5 typical scientific domains: biology, physics, chemistry, geography, and mathematics. Specifically, we construct 3 types of Q-C-A pairs, for evaluating the complicated real world reasoning capabilities of rerankers towards scientific domains. *Noisy Contexts (NC)* are designed to evaluating the discernability of rerankers towards noisy contexts. *Semantically Similar but Logically Irrelevant Contexts (SSLI)* are constructed to evaluating the reasoning ability of rerankers towards similar but do not supporting correct answer contexts. *Counterfactual Contexts (CC)* are built to evaluate the ability of rerankers when facing factually incorrect or contradictory knowledge. Finally, we running 13 popular rerankers on 11 LLMs. Experimental results reveal that rerankers can significantly improves performance in RAG-LLMs, with cross-encoder architectures offering the strongest gains. However, in more complicated multi step reasoning tasks, the quality of final answers heavily depends on LLMs' internal reasoning capacity. **We summarize our contributions as follows**:

- To the best of our knowledge, this work is the first benchmark designed for evaluating rerankers in RAG-LLMs.
- We construct 4 types of diagnostic datasets to deeply assess the capabilities of rerankers towards scientific domains.
- Through evaluating popular 13 rerankers on 5 families of LLMs, we summarize several key observations, which reveal their advantages and limitations.

2 Related Work

To position our work in the context of existing research, we review prior efforts across three key areas: scientific QA datasets, benchmarks for scientific QA tasks, and reranking methods that have been explored in RAG-based LLM systems

2.1 Scientific Question Answering Datasets

Existing QA datasets can be broadly divided into two categories: open-domain and scientific datasets. In open domain, Natural Questions (NQ) [24], TriviaQA [23], and WebQuestions [49] have served as popular open-domain QA datasets. Furthermore, HotpotQA [54] and MuSiQue [43] emphasize multi-hop reasoning and complex context synthesis. Scientific literature plays a critical role in advancing human knowledge, and accurate comprehension and retrieval of scientific documents is essential for research progress [20]. In the biomedical domain, BioASQ [44] are generated based on PubMed abstracts, with expert annotated factoid and list type questions. PubMedQA [19] further extends this by focusing on yes/no/maybe questions derived from clinical research papers. MATH [13] is designed to assess the procedural reasoning abilities of LLMs across various mathematical domains. AquaRat [25] offers algebra word problems with step by step rationales. OpenBookQA [29] and ARC [7] are general scientific QA datasets. They construct science exam style questions at elementary and high school levels, emphasizing both retrieval and reasoning.

Sorts of studies [1, 18] shown that question related context, i.e., external knowledge or facts, can help LLM generate more reliable and accurate answers. However, except for PubMedQA and BioASQ, most datasets do not provide a large amount of contextual information.They assume that relevant context is already retrieved and do not simulate real world conditions where the model must reason over partially relevant, noisy, or even misleading documents. This limits their ability to reveal the nuanced performance of retrieval and rerankers.

2.2 Scientific Question Answering Benchmarks

Scientific QA has become a popular use case for evaluating LLMs due to its demand for factual precision and deep domain understanding. Several recent benchmarks have been proposed to test LLMs on scientific reasoning tasks. Similarly, APBench [50] focuses exclusively on astrodynamics, evaluating models on real world

Table 1: Comparison of SciRerankBench among related datasets, toolkits and benchmarks.

Type	Work	Source Data	Explainability	Scientific Field	Knowledge Level	Rerank Oriented	Dataset Size
Dataset	NQ [24]	Google search log	✗	✗	General	✗	307k (train), 7.8k (dev/test)
	HotpotQA [54]	Wikipedia	✓	✗	General	✗	113k
	PubMedQA [19]	PubMed abstracts	✓	Bio.	Expert	✗	1k expert, 61k unlabeled, 211k generated
Toolkit	FlashRAG [18]	Public benchmarks	✓	✗	General	✓	unknown
	rankify [1]	Public benchmarks	✓	✗	General	✓	unknown
Bench.	CRAG [53]	Web search log	✓	✗	General	✗	4.4K
	AGIEval [61]	Exams	✗	Geo., Bio., His.,Chem., Phy., Eng., Chi., Math, Law, Log.	High school	✗	8k
	SciEval [40]	Public datasets	✗	Phy., Chem., Bio.	College	✗	15k
	SciBench [48]	University exams	✓	Phy., Chem., Math	University	✗	2.1k
	PHYBench [36]	Physic exams	✓	Phy.	High School, College, Olympiad	✗	500
	SciHorizon [35]	Public datasets	✓	Math., Phys., Chem. Bio., Earth & Space	College	✗	unknown
	SciRepEval [39]	Scientific literature	✗	20 subjects	Expert	✗	12.4M
	Ours	Scientific literature	✓	Phy., Chem., Bio., Geo., Math.	Expert	✓	4.5k

aerospace tasks. ChemBench [30] benchmarks LLMs on diverse chemistry tasks, evaluating both factual knowledge and reasoning. PHYBench [36] focuses on physical reasoning through tasks like dimensional analysis and estimation. However, these benchmarks are typically limited to a single scientific domain, which restricts their ability to evaluate the generalizability and robustness of LLMs across diverse fields of scientific inquiry.

To overcome the limitations of domain specific benchmarks, several multidisciplinary QA benchmarks have been proposed to evaluate LLMs across a broader spectrum of scientific fields. SciEval [40] focuses on a comprehensive multi axis evaluation framework, while SciBench [48] pushes the level of questions to a university setting and incorporates multimodal prompts. AGIEval [61] expands the evaluation of LLMs beyond factual recall, incorporating high difficulty tasks across multiple disciplines , many of which are based on real world human exams. SciRepEval [39] introduces a unified benchmark comprising 25 diverse to evaluate the generalization ability of scientific document representation models across multiple task formats. SciHorizon [35] provides a comprehensive benchmark to evaluate the AI-for-Science readiness of both scientific datasets and LLMs. Rankify [1] and FlashRAG [18] both provide a comprehensive and modular Python toolkit for retrieval, reranking, and generation, supporting reproducible experimentation and systematic benchmarking. While SciEval, SciBench, AGIEval, and SciRepEval offer broad and diverse evaluation settings, they do not isolate or assess the rerankers critical to RAG pipelines. Conversely, Rankify and FlashRAG provide infrastructure for evaluating rerankers, but their datasets remain largely grounded in general domains rather than the scientific literature.

2.3 Rerankers in Retrieval-Augmented Generated LLMs

Rerankers in RAG-LLMs can be categorized into the following architectures. *Dense cross-encoder* rerankers: BGE [6], Jina [21], BCE [31], and MXBAI [38]. They compute contextualized relevance scores via bi-sequence encoders and have demonstrated strong performance on various ranking tasks. *Sparse lexical* rerankers: such as SPLADE [11] utilize term level matching with sparsity constraints to maintain interpretability and alignment with classical information retrieval principles. Lightweight alternatives like MiniLM [47] leverage distilled self-attention mechanisms to reduce inference cost while maintaining reasonable accuracy. *Late-interaction* rerankers like ColBert [37] delay token level interactions until the scoring stage, offering an efficiency accuracy trade-off suitable for large scale retrieval. More recently, *LLM-based* rerankers, such as RankGPT [41], LLM2Vec [4] exploit the reasoning capabilities of large generative models to perform zero-shot or instruction guided reranking, along with increased computational overhead. *Sequence-to-sequence* rerankers, such as In-Rank [60], tokenize query-document pairs and employ generative decoding to assess relevance, making them suitable for instruction driven reranking while not relying on LLMs. Additionally, *listwise ranking models*: RankT5 [63], ListT5 [55], adopt groupwise learning objectives to optimize the global ordering of candidate contexts. Twolar [2] proposes a two-step LLM augmented distillation framework for passage reranking, achieving strong performance by transferring knowledge from powerful LLMs to lightweight student models. Rank-R1 [64] enhances the reasoning capability of document rerankers by introducing a reinforcement learning framework optimized for

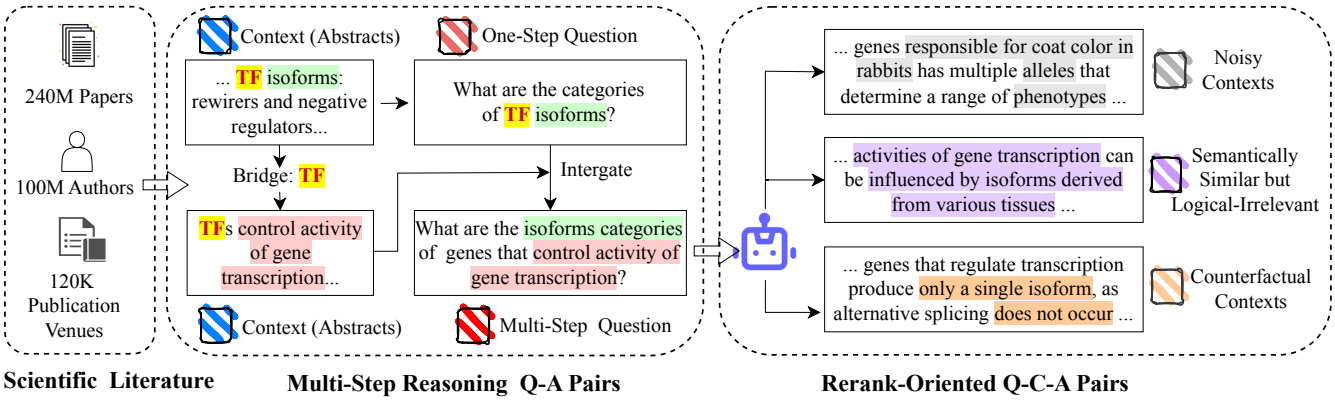


Figure 2: Overview of the synthetic dataset process. We first generate multi-hop reasoning and diverse types of distractors to simulate real world complicated retrieved contexts. Distractors include *Noisy Contexts*,*Semantically Similar but Logically Irrelevant Contexts* and *Counterfactual Contexts*.

answer aware rewards. It directly optimizes reranking performance based on downstream QA accuracy rather than traditional relevance labels. G-RAG [8], a *graph based reranker* that models semantic and relational connections between retrieved documents using graph neural networks (GNNs). *Agent-based rerankers* explicitly formulate the reranking process as a decision making task. Rearank [58] introduces a reasoning agent trained via reinforcement learning to iteratively examine candidate passages and generate a globally optimal ranking.

However, existing benchmarks do not evaluate these rerankers independently in a reliable way. They are usually designed to evaluate end-to-end RAG systems and assume that the retrieved contexts are already of high quality. This makes it difficult to assess how well a reranker can distinguish relevant from irrelevant or misleading passages, especially in scientific tasks where accuracy, terminology, and reasoning precision are crucial.

3 Reranking-Oriented Benchmarking Dataset

This section outlines our data construction process, including source data collection, multi-step QA generation, and the synthesis of reranking-oriented question-context-answer (Q-C-A) pairs.

3.1 Scientific Literature Source Data

We have collected a large number of scientific papers from open access academic repositories [33]. The selected papers span multiple disciplines, including computer science, biology, physics, and materials science, ensuring coverage of a diverse range of scientific terminologies. Each paper contains structured metadata (title, abstract, authors, publication year) and unstructured content (abstracts and full texts). These information is used in 2 ways: generate QA pairs and serve as the RAG retrieval corpus. Our dataset covers 5 major disciplines: physics, chemistry, biology, geography, and mathematics, ensuring broad topical diversity and scientific relevance. All records were imported into a vector database [34], enabling efficient dense retrieval. The abstract serves as the primary source for both question generation and retrieval context.

3.2 Synthetic Multi-Step Reasoning Q-A Pairs

To construct QA pairs from scientific literature, we apply a dual-method approach that includes both single-hop and multi-hop question generation techniques. This ensures a diverse set of questions ranging from basic fact extraction to multi-step reasoning. For single-hop question generation, we adopt the *LMQG* [45]framework, a language model based method to generate natural language questions from input passages. Specifically, we use the unsupervised mode of LMQG, where only the abstract text is required as input. This approach allows for large scale, automatic generation of factoid style questions from scientific abstracts with minimal human intervention. For multi-hop question generation, we adopt the *Unsupervised-Multi-hop-QA* [32] framework to construct questions that require reasoning across 2 semantically linked contexts.The overall process is illustrated in Figure 2. Specifically, for each source abstract in our scientific papers, we retrieve the most semantically similar abstract based on dense embeddings. The resulting pair comprising the original abstract and its nearest neighbor—is concatenated and fed into the multi-hop question generation model. This setup are designed to generate questions that require integrating information across 2 distinct but related scientific papers, simulating real world complicated scientific reasoning scenarios.

3.3 Synthetic Reranking-Oriented Q-C-A Pairs

Fang et al. [10] analyzed how different types of noisy retrieved texts affect the performance of LLMs. They identified 3 common noise types: contexts that are superficially related to the query but lack the correct answer,contexts that are irrelevant to the query , and contexts that are topically related to the query but contain incorrect information. Their findings show that these types of context can easily confuse large language models. To evaluate the ability of rerankers towards scientific domains, we construct the reranking-oriented Q-C-A pairs in this paper. As illustrated in Figure 2, we construct 3 types of disruptive contexts:

Noisy Contexts (NC): A common challenge in real-world scientific retrieval scenarios is the inevitable presence of irrelevant or unrelated passages mixed in with relevant contents [24, 25, 30, 40].

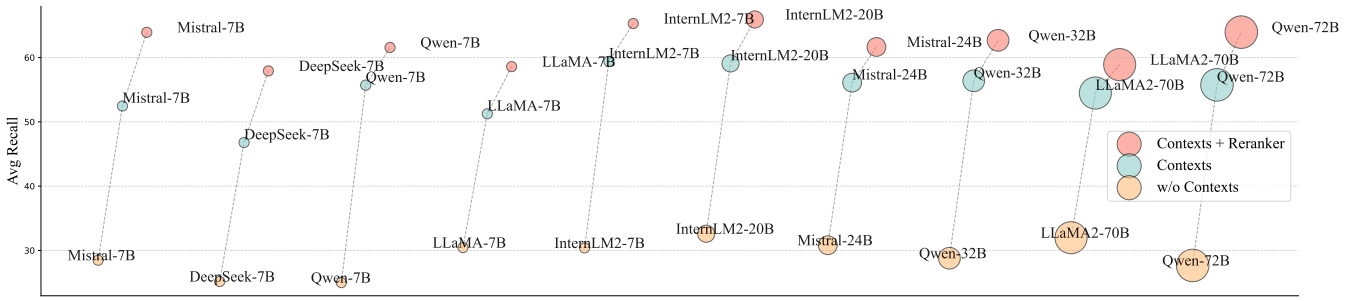


Figure 3: Evaluation results under 3 different settings: without any context, with retrieved contexts, and with contexts after applying the reranking stage by recall scores.

To simulate this practical challenge and evaluate the ability of rerankers to distinguish relevant information within noisy data, we design the Noisy Contexts setting. Each query in this setting is paired with 5 passages confirmed to be relevant, mixed with 95 randomly sampled unrelated passages. This setup serves as a fundamental test for assessing the discernability of rerankers towards noisy contexts.

Semantically Similar but Logically Irrelevant Contexts (SSLI): Passages that share high semantic similarity with the query but are logically irrelevant often appear as challenge. Such contexts can easily mislead rerankers [19, 36], as they match surface level keywords and phrases, yet do not contain the necessary information to answer the question. This mismatch between semantic similarity and logical relevance requires rerankers to possess strong reasoning and contextual understanding abilities. To simulate this challenge and rigorously evaluate rerankers' reasoning skills, we design the Semantically Similar but Logically Irrelevant setting. Each query is paired with 90 standard candidate passages and 10 passages that are semantically similar but logically irrelevant. This setup aims to test the model's capacity for fine grained logical discrimination, going beyond simple semantic matching to accurately identify answer bearing content.

Counterfactual Contexts (CC):Retrieval passages may contain information that is factually incorrect or contradictory to established knowledge [1, 35, 39, 50]. Such passages pose a significant challenge because they can mislead rerankers relying solely on semantic similarity, requiring models to assess not just relevance but also factual correctness. To simulate this challenge and rigorously evaluate rerankers' counterfactual reasoning abilities, we design the Counterfactual Contexts setting. Each question includes 90 standard candidate passages and 10 passages containing plausible yet factually incorrect or contradictory information. This setup tests the model's ability to discern truthfulness and accuracy in addition to semantic alignment.This setting presents a more realistic and stringent scenario for scientific QA systems, where distinguishing between fact and fiction is essential for trustworthy ouputs.

4 Experiment

We evaluate 13 widely used rerankers on the SciRerankBench benchmark, covering their effectiveness, robustness, reasoning capability, and efficiency. The following subsections describe our experimental setup and detailed results across various evaluation dimensions.

4.1 Experimental Settings

Rerankers. We categorize all rerank baselines into 8 groups based on their architectures. (1) *Dense* rerankers: BGE [6], BCE [31], Jina [21]and MXBAI [38]. (2) *Sparse lexical* rerankers: SPLADE [11]. (3) *Listwise* rerankers: RankT5 [63] and ListT5 [55]. (4) *Seq2Seq* rerankers: In-Rank [60]. (5) *Lightweight student* rerankers: Twolar[2] and MiniLM[47]. (6) *Late-interaction* rerankers: ColBert[37] (7) *LLM-based* rerankers: RankGPT [41],LLM2Vec[4] (8) *Agent-based* rerankers: Rearank [58] [1].

LLMs. In our experiments, we evaluate a diverse set of 11 open-source LLMs drawn from different families, as summarized in Table 3 in the Appendix. Mistral models (7B and 24B) [17] incorporate architectural enhancements such as *Sliding Window Attention (SWA)* and *Grouped-Query Attention (GQA)* to improve long-context modeling and computational efficiency. The LLaMA2-70B model [42] similarly employs GQA along with *Rotary Positional Embeddings (RoPE)* to better handle extended input sequences. The Qwen model [52] uses the standard *multi-head attention (MHA)* mechanism, combined with GQA and RoPE, to support multi-language and long-context input while maintaining performance. DeepSeek-V3 [26] introduces a sparse attention mechanism, *mixed experts (MoE)* and *multi-layer attention (MLA)* to improve large-scale reasoning efficiency and knowledge capacity.We include InternLM2.5 [5], a multilingual, open-weight transformer model supporting long-context understanding (up to 200K tokens) and optimized for both instruction-following and reasoning tasks.

Evaluation Protocols and Metrics. To comprehensively evaluate model performance, we design a two-level evaluation strategy targeting both rerankers and LLMs. For rerankers, we compare the top contexts against annotated relevance labels to assess their ability to retrieve the most informative and factually correct passages for a given scientific question. *Recall@10* evaluate how many of the known relevant passages are ranked within the 10, respectively, thus testing retrieval completeness. For LLMs, we assess their ability to generate accurate and complete answers by comparing outputs to golden answers. We use a token level metrics: *Recall*, which captures how completely the model reproduces the reference content.

[1]As the codes of some rerankers, i.e., Rank-R1 [64] and G-RAG [8], are unavailable, we do not evaluate these models. Besides, the estimated time of evaluating RankGPT would be approximately 1,611 hours. Thus we do not run RankGPT [41] in this paper.

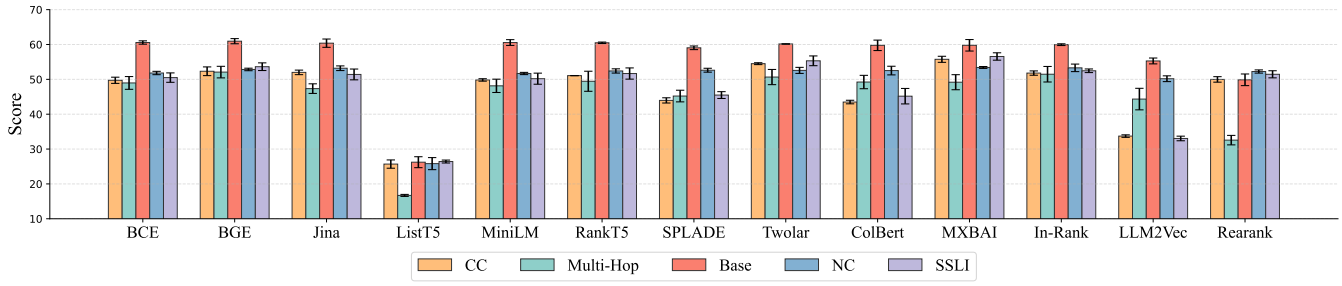


Figure 4: Evaluation of rerankers on five tasks using *Recall* metric

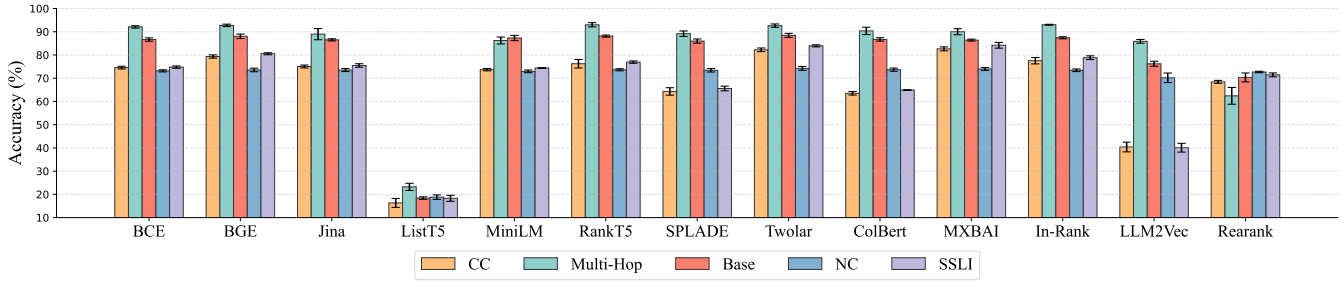


Figure 5: Evaluation of rerankers on five tasks using *Recall@10* metric.

This evaluation framework allows us to analyze the individual contribution and limitations of rerankers versus generation models in the RAG pipeline, especially under challenging scientific question answering scenarios. Definitions of these metrics are provided in Appendix A.2. This two-level evaluation helps us figure out whether mistakes come from retrieving the wrong information, or from the model not using the information well when generating answers. By looking at both parts separately, we can better understand where the system needs improvement. This is especially important for handling complex scientific questions where both accurate retrieval and strong reasoning matter.This approach also helps compare different system components fairly, making it easier to identify which part contributes most to the overall performance.

Implementation Details. The rerankers are conducted on 4 NVIDIA A100 GPUs. All LLMs and rerankers are used with their default configurations. We conduct 3 independent experiments to ensure the reliability and stability of our evaluation results. To ensure a fair comparison, all models are evaluated under a zero-shot setting, without task-specific fine-tuning or additional adaptations. We collect 100,000 scientific articles and store their abstracts in a Qdrant vector database [34] to facilitate dense retrieval for RAG-based systems. Then we preprocess and filter the abstract texts to ensure quality and consistency. Specifically, we retained only those abstracts with a length between 100 and 500 characters. This range was selected to exclude low information or overly terse abstracts, e.g., placeholder texts, while also avoiding excessively long passages that could complicate automatic question generation or overload retrieval models. Following previous works [1, 18, 56], 100 candidate contexts are first retrieved with BGE as the retriever, then the top 10 are selected by rerankers, which are used as input to LLMs for final answer generation. In addition, a sampling test dataset showed that RankGPT required 3.5 hours to process 100 samples. Based on this, the estimated time to complete the evaluation on the full dataset would be approximately 1,611 hours. Due to this high computational cost, we do not running RankGPT in this paper.

4.2 Evaluating the Usefulness of Rerankers in RAG Pipelines.

To evaluate the impact of context retrieval and reranking on answer generation quality, we compare LLM performance across 3 distinct settings: (1) zero-context generation (no retrieved context), (2) naive retrieval (Top-10 dense retrieval without reranking), and (3) RAG with reranked contexts (Top-10 after reranking). As shown in Figure 3, all architectures benefit significantly from adding external contexts, with contexts alone contributing 20–30 point gains in recall on average. Among all families, InternLM and Qwen exhibit the strongest improvements from reranking, suggesting that these models better leverage high quality contexts when ranked effectively. InternLM2-20B consistently achieves the highest recall across all settings, while Qwen models also demonstrate robust gains, likely aided by their multilingual pretraining and long-context optimization. In contrast, DeepSeek models show moderate performance without context but respond well to retrieval, albeit with limited further gains from reranking.

Takeaways.

- *While all LLMs benefit from retrievers and rerankers, the ability to exploit high quality reranked context varies across different families and architectures.*
- *After reranking the context with rerankers, InternLM and Qwen achieve the best performance.*

4.3 Evaluating the Discernability of Rerankers towards Noisy Contexts

In the *NC* dataset, where 5 relevant contexts are mixed with 95 random distractors. We observe that most rerankers are able to retrieve nearly all relevant contexts within the top-10 results, as indicated by the high recall@10 scores across most evaluated models. The presence of a large number of unrelated contexts appears to disperse model attention, impairing the models' ability to maintain coherent focus on relevant information. Even models with strong multi context reasoning capabilities often struggle to prioritize truly important passages over distractors. As a result, the effective recall of useful information during answer generation degrades, leading to lower final answer quality. Furthermore, when irrelevant contexts are intermixed, models may mistakenly rely on misleading information during both retrieval and generation stages. This phenomenon highlights the necessity of not only retrieving relevant documents but also ensuring high contextual purity. Doing so helps avoid attention diffusion and knowledge contamination in scientific question answering pipelines.

Takeaways.

- *LLMs often struggle to effectively filter out irrelevant contexts when handling tasks with a large amount of unrelated information, which affects the quality of final answers.*
- *Effective scientific QA requires both precise retrieval of relevant information and minimizing the influence of irrelevant content.*

4.4 Evaluating the Reasoning Ability of Rerankers towards Semantically Similar but Logical-Irrelevant Contexts

The *SSLI* dataset focus on evaluating rerankers' ability to handle semantically challenging cases, where contexts are semantically similar yet unanswerable. In these complex scenarios, cross-encoder architectures demonstrate significant advantages. By jointly encoding the query and document, cross-encoders allow fine-grained feature interactions, capturing subtle semantic differences crucial for distinguishing between true and misleading information. In particular, MXBAI achieves the highest recall scores across nearly all evaluation settings, highlighting the strength of cross-encoder designs in complex reasoning tasks.In contrast, sparse retrieval models, which rely on masked language model objectives to produce sparse representations, exhibit significantly lower recall (approximately 44%). Although sparse models offer faster inference speed—achieving roughly 71.4% the inference time of dense architectures—their limited semantic granularity hinders their ability to identify fine-grained factual inconsistencies, leading to notable performance degradation on nuanced reasoning tasks. Late interaction models such as ColBert, which independently encode queries and documents before performing token wise interaction, maintain strong recall in random distractor settings but struggle when deeper semantic comprehension is required. ColBert's recall drops to 43.47% on counterfactual tasks and 45.17% on semantic confusion tasks, illustrating the limitations of independent representation without full context fusion. LLM-based embedding methods like LLM2Vec underperform on semantically challenging tasks, scoring only 33.72% on counterfactual and 33.04% on semantic confusion. Although it leverages frozen LLMs for dense representation, the lack of joint encoding limits its ability to capture subtle semantic mismatches crucial for accurate reranking.

Takeaways.

- *Cross-encoders outperform other rerankers on semantically challenging tasks, due to their fine-grained query-document interaction.*

4.5 Evaluating the Ranking Ability of Rerankers

To evaluate the effectiveness of rerankers independently of final answer generation, we additionally present recall@10 to directly measure how well rerankers prioritize relevant information in the top-ranked results.For Recall@10, most rerankers achieve relatively high scores across datasets, including multi-hop and complex QA tasks. However, a discrepancy emerges when examining the final answer quality: despite high Recall@10, the generated answers from LLMs are often of lower quality compared to those from the base retrieval-only setup. This indicates that, while rerankers are effective in ensuring that relevant contexts are included within the top-10 candidates, the ultimate answer quality remains constrained by the LLM's own generation capabilities. The finding highlights a limitation where retrieval improvements alone cannot fully compensate for deficiencies in the LLM's ability to synthesize and reason over retrieved knowledge. In multi-hop tasks, rerankers can successfully retrieve sufficient relevant context. However, even with ample context, LLMs fail to achieve expected performance due to inherent limitations in their reasoning capabilities, which ultimately restrict the quality of final answers. Beyond limitations in LLM generation, differences in reranker design also contribute to performance gaps across tasks. Although REARANK uses a listwise reinforcement learning framework with local window-based ranking over sliding windows, its architecture lacks global cross-document encoding or token-level interaction. As a result, it cannot effectively aggregate complementary evidence scattered across passages, leading to lower Recall@10 (62.41) on multi-hop reasoning tasks.

Takeaways.

- *For multi-hop tasks, retrieval and rerankers are able to select high-relevant contexts, but the final performance still depends on the inherent reasoning capacity of LLMs.*
- *Reranker architectures also influence performance on muti-hop task.*

4.6 Evaluating the Running Efficiency of Rerankers

This section analyze the running efficiency. The average inference time of one question is shown in Figure 6. We observe clear running efficiency differences across reranker architectures. Sparse models like SPLADE achieve the fastest inference (0.49s), owing to their non-interactive, inverted-index design. Lightweight sentence encoders such as MiniLM also offer rapid performance (1.34s), making them attractive for latency-sensitive applications. In contrast, cross-encoder models demonstrate a trade-off: BCE and BGE are relatively efficient (3.5s), while deeper or more expressive models

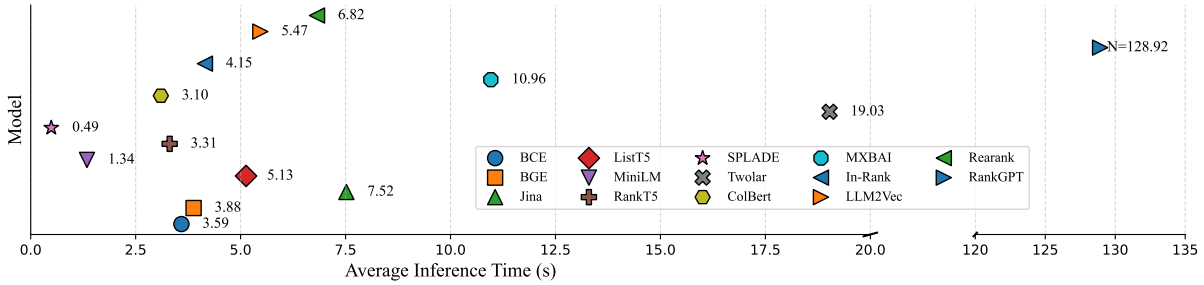


Figure 6: Running efficiency analysis of rerankers.

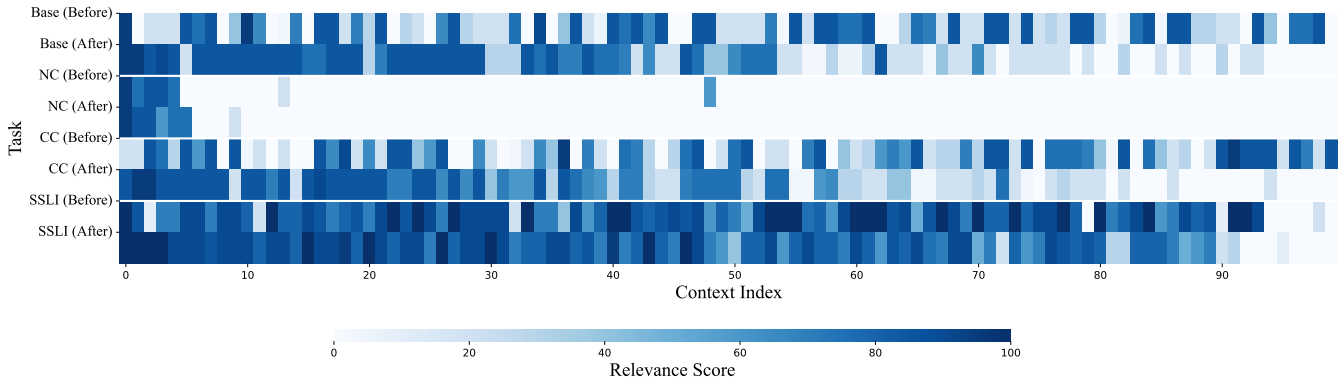


Figure 7: Visualization of question-context relevance (before and after rerankering) under 4 evaluation tasks.

like MXBAI incur higher cost (10.96s) due to full query-document interaction. Twolar emphasizes ranking accuracy through a two-stage, listwise-aware architecture. Thus, it takes more time than others. LLM2Vec uses frozen LLMs to compute dense vector similarities without decoding, resulting in a moderate inference time of 5.47s due to representation extraction and comparison. Rearank adopts an agent-based framework with iterative LLM reasoning, leading to higher latency (6.82s) mainly from multi-round document evaluation and reordering. RankGPT leverages the reasoning power of large language models for reranking through a generative, prompt-based approach. Its extremely slow inference speed, which is caused by having to process each candidate document one at a time through a LLM.

Takeaways.

- *Better reranking performance often comes at the cost of increased inference time.*

4.7 Question-Context Relevance Under Four Task Settings

To better understand the effectiveness of the rerankers in distinguishing relevant from irrelevant contexts, we visualize the context ranking distributions before and after reranking in 4 tasks. We use DeepSeek-V3 671B to assess the semantic relevance between contexts and the given question. The visualization results of the relevance heatmap are shown in Figure 7. In the heatmap visualization, we group the results of the same task into pairs, allowing us to observe changes in color intensity to evaluate the effectiveness of the reranking. An overview of the 4 tasks reveals that after reranking, the relevant contexts are clearly moved to the top positions across all tasks. Particularly in the NC task, even when faced with a large amount of irrelevant context, the rerankers is still able to correctly prioritize the relevant contexts. However, in the SSLI and CC tasks, the model incorrectly ranks irrelevant contexts among the top , leading to poor performance in LLM as discussed in Section 4.4. This suggests that the rerankers limitations in its ability to perform fine-grained semantic reasoning beyond surface-level relevance.

Takeaways.

- *Rerankers effectively promote relevant contexts to higher positions across multiple tasks, especially in noisy settings, but still struggle in scenarios that require deep semantic understanding beyond surface-level similarity.*

5 Conclusion

In this paper, we introduce SciRerankBench. To the best of our knowledge, it is the first publicly available benchmark for evaluating rerankers in RAG-LLMs towards scientific domains. We design datasets covering noisy contexts, semantically similar but logically irrelevant contexts and counterfactual contexts comprehensively evaluate the rerankers. Evaluating 11 rerankers and 11 LLMs on these tasks, our experiments show that while rerankers struggle to filter logically irrelevant yet semantically similar passages. Moreover, final answer quality remains constrained by the inherent reasoning limitations of current LLMs.

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A Appendix

A.1 Detailed Dataset Construction Information

Source Literature. We choose OpenAlex [33] as our source data to construct our synthetic scientific questions. OpenAlex is an open, comprehensive scholarly metadata platform that provides structured information on research papers, authors, institutions, concepts, and more. This database contains over 250 million works, more than 100 million unique author profiles, and metadata for approximately 120,000 publication venues, including both journals and conferences.

Dataset Construction Pseudocode. Algorithm 1 outlines the dataset construction pipeline, including abstract preprocessing, single- and multi-hop QA pair generation, and the creation of three datasets which designed to evaluate different challenges.

Algorithm 1: Dataset Construction Process

Input: Source scientific corpus C
Output: QA Pairs D_{QA}, Reranking Set D_{rank}

// **Step 1: Preprocessing & Indexing**;
$C' \leftarrow$ Filter abstracts: len $\in [100, 500]$;
$V \leftarrow E_{bge}(C')$;
Insert(V into Qdrant index I);
// **Step 2: QA Generation**;
foreach $a \in C'$ **do**
  $(q_1, ans_1) \leftarrow \text{LMQG}(a)$;
  Add (q_1, ans_1) to D_{QA}^{single};
  $a' \leftarrow \text{NearestNeighbor}(a, I)$;
  if $a' \neq NULL$ **then**
    $(q_2, ans_2) \leftarrow \text{UMQG}(a, a')$;
    Add (q_2, ans_2) to D_{QA}^{multi};
// **Step 3: Construct Reranking Subsets**;
foreach $(q, ans) \in D_{QA}^{single}$ **do**
  $P \leftarrow \text{DenseRetrieve}(q, I, k = 100)$;
  $P_{rand} \leftarrow \text{Mix}(5 \cdot \text{Rel}, 95 \cdot \text{Random})$;
  $P_{conf} \leftarrow \text{InjectConfusers}(P, 10)$;
  $P_{cfact} \leftarrow \text{InjectCounterfacts}(P, 10)$;
  Add (q, P_{rand}, ans) to D_{rank};
  Add (q, P_{conf}, ans) to D_{rank};
  Add (q, P_{cfact}, ans) to D_{rank};
// **Step 4: Final Evaluation Set**;
$D_{\text{eval}} \leftarrow D_{rank} \cup D_{QA}^{multi}$;

Specific Dataset Instances/Samples. Here we provide a specific dataset instances or sample, which comprising a scientific question, its corresponding golden answer, and a set of 100 contextual passages. The sample is shown in the Figure 9 to Figure 11.

Dataset Statistics. Table 2 provides a detailed breakdown of the number of QA instances across five scientific subjects—Biology, Math, Physics, Geology, and Chemistry—under five distinct task types. These task types correspond to the benchmark datasets described previously: NC,Base,CC,Multi-hop,SSLI.

Table 2: QA Counts for different subjects and tasks.

Subject	Task Type	Number
Biology	NC	2499
	Base	2499
	CC	2496
	Multi-Hop	1246
	SSLI	2497
Math	NC	2494
	Base	2494
	CC	2491
	Multi-Hop	1631
	SSLI	2493
Physics	NC	2491
	Base	2491
	CC	2494
	Multi-Hop	1425
	SSLI	2492
Geology	NC	2493
	Base	2493
	CC	2493
	Multi-Hop	1598
	SSLI	2496
Chemistry	NC	2499
	Base	2499
	CC	2497
	Multi-Hop	1087
	SSLI	2498

A.2 Detailed Experimental Settings

Detailed LLM Information. Detailed information of LLM models evaluated in this paper is shown in Table 3. These models are from 5 LLM families, i.e., Mistral, LLaMA, DeepSeek, Qwen and InternLM2 , with parameter sizes ranging from 7B to 72B+.This diverse set of LLMs also enables us to introduce a new evaluation perspective—using the quality of LLM-generated answers as an indicator for how well a reranker supports real task performance.

Detailed Reranker Information. Detailed information of the rerankers is shown in Table 4.

Prompting Template. To standardize the evaluation of reranking models, we adopt a unified prompting template that guides models to generate concise answers based solely on the provided contexts. As illustrated in Figure 8, the prompt explicitly instructs the model to refrain from using external knowledge and to return "No Answer Present" when no relevant information is found.

Detailed Information of Evaluation Protocols and Metrics. For rerankers, we compare the top-ranked contexts against annotated relevance labels to assess their ability to retrieve the most informative and factually correct passages for a given scientific question. *Recall@10* evaluate how many of the known relevant passages are

Table 3: LLMs evaluated in this paper.

Model	Family	Size
Mistral-7B-v0.2	Mistral	7B
Mistral-24B	Mistral	24B
LLaMA-7B	Meta (LLaMA)	7B
LLaMA2-70B	Meta (LLaMA2)	70B
DeepSeek-V3-7B	DeepSeek	7B
DeepSeek-V3-671B	DeepSeek	671B
Qwen-7B	Qwen	7B
Qwen-32B	Qwen	32B
Qwen-72B	Qwen	72B
InternLM2-7B	InternLM2	7B
InternLM2-20B	InternLM2	20B

Table 4: RAG reranking methods evaluated in this paper.

Model	Category	Reranking Strategy
BGE	Transformer	Cross-Encoder
Jina	Transformer	Cross-Encoder
BCE	Transformer	Cross-Encoder
MXBAI	Transformer	Cross-Encoder
MiniLM	Sentence Transformer	distilled self-attention
ColBert	BERT	late interaction mechanism
In-Rank	Seq2Seq model	tokenizes query-document pairs
SPLADE	Sparse	MLMs
RankT5	T5-based	ranking losses
ListT5	T5-based	m-ary tournament sort
Twolar	T5-based	two-step distillation approach
LLM2Vec	LLM-based	LLM Embedding Similarity
RankGPT	LLM-based	sliding window
Rearank	Agent-based	reinforcement learning

Our Prompting Template

Instructions:
You are given a question and a set of contexts. Your task is to provide a clear and concise answer to the question based on the contexts provided.
Answer based **only** on the contexts. If none are relevant, say: **No Answer Present.**
Keep your answer short — ideally within 2 sentences. Do not include references or restate the question.
—
QUESTION: {QUESTION}
CONTEXTS: {CONTEXTS}
###
ANSWER:

Figure 8: Prompt used in the Rerank Evaluation Task

ranked within the 10, respectively, thus testing retrieval completeness. For LLMs, we assess their ability to generate accurate and complete answers by comparing outputs to gold-standard answers. We use a token-level metrics:*Recall*, which captures how completely the model reproduces the reference content.

Recall is computed using tokenized outputs. Let P and T be the predicted and true token sets. Then:

$$\text{Recall} = \frac{|P \cap T|}{|T|} \times 100,$$

This metrics reflect the lexical similarity between the predicted answer and the ground truth, helping to quantify completeness in LLM-generated outputs.

Contain Answer Score evaluates whether the gold answer content is covered within the retrieved context passage. This is a soft semantic measure, defined as:

$$\text{ContainAnswer}(A, C) = \frac{|\text{tokens}(A) \cap \text{tokens}(C)|}{|\text{tokens}(A)|}$$

If the overlap exceeds a predefined threshold (e.g., 0.6), the passage is considered to contain the answer. This metric is particularly useful for evaluating context sufficiency in open-domain and scientific QA.

Recall@10 compute how many relevant contexts are successfully retrieved within the top-k results:

$$\text{Recall@}k = \frac{\text{\# relevant items in top } k}{\text{Total relevant items}} \times 100$$

Table 5: Performance of LLM-based and Agent-based rerankers across various evaluation tasks

Reranker	Subject	Multi-Hop	NC	CC	SSLI	Base
LLM2Vec	Bio.	33.72±0.36	32.77±0.45	31.78±0.52	31.34±0.61	31.46±0.50
	Geo.	44.34±3.09	41.84±2.70	38.45±2.10	37.06±1.85	35.99±1.75
	Chem.	55.28±0.84	53.75±0.88	51.35±1.10	49.39±1.05	49.45±0.90
	Phy.	50.18±0.83	47.82±0.75	44.42±0.82	43.96±0.65	45.69±0.88
	Math.	33.04±0.66	29.49±1.10	31.05±0.92	29.26±1.15	28.99±1.90
Rearank	Bio.	49.98±0.79	48.57±1.27	47.11±0.81	46.48±2.16	46.69±0.94
	Geo.	32.56±1.36	30.73±1.92	26.71±2.38	24.83±0.84	23.48±1.17
	Chem.	49.87±1.67	48.51±0.88	46.32±1.21	44.56±1.18	44.64±0.61
	Phy.	52.25±0.46	49.76±0.23	46.27±0.70	45.86±0.32	47.64±0.87
	Math.	51.45±1.02	45.90±1.34	48.35±0.50	45.62±1.23	45.22±3.21

A.3 Detailed Experimental Results

Detailed main results are shown in Table 6 and Table 7. These tables report the performance metrics of various rerankers across multiple tasks and subjects under two different backbone LLMs, Qwen and LLaMA. The comparison highlights the general effectiveness and robustness of the rerankers, with only minor performance variations across subjects, suggesting that the models generalize well across domains. In addition, Table 5 compares two representative reranking approaches: the LLM-based LLM2Vec and the agent-based Rearank. This table offers a more focused analysis of how rerankers with different architectural designs and inference paradigms perform under the same evaluation settings. Together, these results provide a comprehensive view of reranker behavior across model types, task types, and subject domains, serving as a reference for future system design and benchmarking.

Table 6: Performance of different rerankers across various evaluation tasks (Qwen-70B)

Reranker	Subject	Multi-Hop	NC	CC	SSLI	Base
BCE	Bio.	55.08±2.62	53.57±2.20	51.66±0.48	51.89±1.40	63.48±0.57
	Geo.	52.44±1.23	51.84±1.53	48.94±1.01	48.14±1.73	60.20±0.42
	Chem.	53.62±2.08	52.65±0.81	49.74±1.22	49.44±2.50	59.61±0.75
	Phy.	49.77±0.90	51.93±1.66	45.32±1.12	46.05±1.45	61.94±0.74
	Math.	52.51±1.48	52.38±0.18	49.26±1.47	47.84±1.52	62.14±2.30
BGE	Bio.	54.55±1.70	53.56±2.00	55.52±0.79	54.56±1.16	64.74±0.71
	Geo.	51.13±0.84	52.25±1.18	52.99±0.59	51.37±1.17	60.91±0.57
	Chem.	52.15±1.62	53.00±0.55	52.88±1.55	51.01±0.89	60.47±1.39
	Phy.	50.45±0.23	52.57±1.47	49.00±1.60	50.37±1.43	61.66±0.73
	Math.	51.45±1.69	52.70±0.29	52.43±0.94	50.92±1.61	62.16±1.63
Jina	Bio.	51.52±1.78	53.89±1.87	53.22±0.83	51.87±1.28	62.85±0.78
	Geo.	47.44±0.95	51.99±1.41	50.92±0.89	49.92±1.36	60.26±0.61
	Chem.	49.74±1.16	52.54±0.73	52.20±1.45	51.97±2.40	59.20±0.79
	Phy.	47.53±1.19	52.01±1.46	48.35±1.62	49.67±2.22	60.33±0.69
	Math.	51.69±1.17	52.76±0.32	52.37±0.54	51.28±1.37	61.30±1.77
ListT5	Bio.	8.89±0.54	26.71±1.28	15.54±1.23	14.69±0.95	15.62±0.60
	Geo.	10.19±2.28	27.20±1.56	15.39±0.56	13.54±1.01	12.83±0.81
	Chem.	8.60±0.72	16.51±0.90	11.21±0.61	2.06±0.32	11.42±0.99
	Phy.	8.80±0.77	20.97±0.95	12.28±0.82	12.01±1.32	10.95±0.85
	Math.	9.17±1.19	28.30±1.46	13.68±0.74	13.41±0.94	12.71±0.89
MiniLM	Bio.	52.95±2.53	53.72±1.92	51.17±1.41	51.29±1.65	63.92±0.94
	Geo.	48.78±0.38	51.75±1.74	48.82±1.30	48.18±2.28	60.29±0.70
	Chem.	50.87±2.57	52.67±0.67	49.22±2.01	48.64±1.78	59.44±0.59
	Phy.	46.71±1.17	52.24±1.70	45.25±0.74	46.55±3.55	60.98±0.66
	Math.	49.64±1.64	52.77±0.69	49.15±0.48	47.83±2.21	61.75±0.50
RankT5	Bio.	14.98±1.77	34.12±1.99	21.08±1.62	20.64±2.36	19.18±0.59
	Geo.	16.78±1.76	34.69±1.10	22.76±1.31	19.44±2.47	19.57±1.55
	Chem.	12.06±1.61	21.89±0.29	17.04±0.63	49.44±0.87	11.32±0.54
	Phy.	14.83±0.78	27.76±1.34	18.48±2.70	18.89±2.50	14.00±1.23
	Math.	18.73±0.76	35.56±0.91	21.57±4.64	20.37±1.78	18.47±0.29
SPLADE	Bio.	14.16±1.12	34.28±2.45	23.40±1.51	19.05±1.26	19.04±0.48
	Geo.	17.27±0.37	34.26±1.30	22.45±1.28	20.65±0.55	19.39±1.10
	Chem.	12.31±1.51	21.84±0.77	18.35±1.95	42.71±3.13	10.66±0.02
	Phy.	14.21±1.25	28.27±1.96	20.49±1.81	21.10±2.76	14.43±0.44
	Math.	19.05±0.53	35.26±0.27	20.39±3.23	20.30±2.48	18.58±0.47
TwoLAR	Bio.	53.69±1.41	53.53±2.04	56.17±0.72	57.24±1.11	63.55±0.84
	Geo.	50.78±0.42	52.05±1.76	55.36±0.96	53.59±1.30	61.12±0.54
	Chem.	52.02±2.02	53.05±0.69	54.20±1.66	53.40±0.85	60.13±0.48
	Phy.	49.57±1.20	52.40±1.70	51.65±0.90	52.19±1.05	61.91±0.37
	Math.	52.10±1.39	53.07±0.47	54.33±0.23	52.90±1.60	62.43±0.41
ColBert	Bio.	53.35±1.96	53.63±1.98	44.92±0.32	44.38±0.74	62.87±0.69
	Geo.	50.35±0.93	51.60±1.75	41.09±0.51	40.27±1.68	60.32±0.26
	Chem.	52.07±1.80	52.82±0.65	42.12±1.04	42.24±1.51	58.96±0.88
	Phy.	49.25±1.09	52.30±1.38	40.30±0.74	39.19±2.83	61.10±0.46
	Math.	51.12±0.79	52.68±0.26	41.97±0.82	41.25±2.11	61.74±0.70
MXBAI	Bio.	53.56±3.25	53.62±1.71	57.18±0.53	58.40±0.96	62.25±0.67
	Geo.	49.26±1.22	51.72±1.48	56.45±0.43	55.40±1.63	59.67±0.64
	Chem.	51.43±1.56	53.05±0.59	54.29±1.12	55.42±1.05	58.83±0.47
	Phy.	48.57±0.74	52.29±1.82	53.33±0.34	54.45±1.89	61.09±0.70
	Math.	50.29±2.11	52.92±0.16	56.61±0.44	54.39±2.41	61.02±0.40
T5	Bio.	54.45±1.46	53.64±1.75	53.92±1.13	53.81±1.17	63.96±0.99
	Geo.	51.08±1.04	51.87±1.64	52.46±0.78	51.40±0.54	60.65±0.06
	Chem.	53.08±1.68	52.76±0.59	52.44±1.87	51.66±0.98	59.67±0.94
	Phy.	50.41±0.86	52.20±1.23	49.11±1.03	48.59±2.62	61.44±0.76
	Math.	52.37±1.38	52.66±0.31	52.95±0.25	51.28±2.12	62.14±0.55

Table 7: Performance of different rerankers across various evaluation tasks (LLaMA2-70B)

Reranker	Subject	Multi-Hop	NC	CC	SSLI	Base
BCE	Bio.	48.97±1.83	51.82±0.48	49.73±0.89	50.50±1.35	60.57±0.48
	Geo.	50.32±1.20	51.94±0.70	47.85±0.95	48.30±1.40	59.12±0.65
	Chem.	50.40±1.15	51.40±0.60	48.92±1.05	49.40±1.20	58.84±0.73
	Phy.	48.36±1.32	51.55±0.88	45.21±0.90	46.82±1.48	61.78±1.12
	Math.	49.55±1.18	51.68±1.12	49.50±1.00	48.96±1.33	60.88±1.40
BGE	Bio.	52.08±1.66	52.84±0.36	52.32±1.25	53.63±1.10	60.95±0.74
	Geo.	51.67±1.45	52.67±0.59	49.35±0.83	51.48±1.18	60.42±0.55
	Chem.	51.74±1.30	52.40±0.64	50.23±1.00	51.92±1.35	60.26±0.88
	Phy.	50.86±1.17	52.55±0.74	47.96±1.08	50.12±1.29	62.20±1.15
	Math.	52.11±1.25	52.69±0.77	51.12±0.91	52.23±1.31	61.03±1.20
Jina	Bio.	47.34±1.38	53.19±0.67	52.00±0.65	51.41±1.55	60.38±1.19
	Geo.	47.67±1.05	52.51±0.74	49.28±1.10	49.21±1.40	59.01±1.30
	Chem.	48.20±1.25	52.13±0.85	50.01±1.03	49.97±1.32	58.32±1.20
	Phy.	46.33±1.36	52.60±0.92	47.39±1.07	47.88±1.45	61.14±1.08
	Math.	47.75±1.12	52.88±0.68	50.23±1.01	49.66±1.44	60.42±1.22
ListT5	Bio.	16.70±0.29	25.82±1.73	25.69±1.19	26.42±0.42	26.23±1.57
	Geo.	10.19±2.28	27.20±1.56	15.39±0.56	13.54±1.01	12.83±0.81
	Chem.	8.60±0.72	16.51±0.90	11.21±0.61	2.06±0.32	11.42±0.99
	Phy.	8.80±0.77	20.97±0.95	12.28±0.82	12.01±1.32	10.95±0.85
	Math.	9.17±1.19	28.30±1.46	13.68±0.74	13.41±0.94	12.71±0.89
MiniLM	Bio.	48.14±1.91	51.68±0.32	49.83±0.35	50.19±1.59	60.56±0.83
	Geo.	49.20±1.45	51.33±0.39	47.55±0.67	48.72±1.51	59.37±0.61
	Chem.	48.97±1.33	51.11±0.45	48.40±0.58	49.45±1.34	59.06±0.78
	Phy.	47.25±1.12	51.26±0.52	45.71±0.62	46.91±1.47	61.34±0.91
	Math.	48.45±1.28	51.38±0.48	48.73±0.59	49.18±1.46	60.61±0.85
RankT5	Bio.	49.45±2.88	52.41±0.61	51.07±0.03	51.33±1.57	60.49±0.21
	Geo.	50.11±2.02	52.10±0.57	48.84±0.72	50.08±1.49	59.73±0.42
	Chem.	49.85±1.89	51.78±0.62	49.71±0.60	50.89±1.40	59.61±0.64
	Phy.	48.23±1.74	52.05±0.68	47.22±0.65	48.98±1.47	61.41±0.72
	Math.	49.35±2.03	52.17±0.66	50.26±0.51	50.70±1.35	60.55±0.85
SPLADE	Bio.	45.21±1.67	52.62±0.56	43.96±0.75	45.38±1.35	59.05±0.52
	Geo.	45.62±1.21	52.38±0.67	42.41±0.89	43.76±1.50	58.30±0.40
	Chem.	45.95±1.00	52.12±0.74	42.85±0.93	44.44±1.42	57.78±0.60
	Phy.	44.16±1.09	52.33±0.59	40.92±0.95	42.88±1.51	60.22±0.73
	Math.	44.90±1.30	52.50±0.62	43.50±0.88	44.48±1.53	59.11±0.90
TwoLAR	Bio.	50.66±2.18	52.57±0.87	54.50±0.27	55.16±0.89	60.15±0.04
	Geo.	51.12±1.55	52.44±0.72	51.58±0.61	52.75±0.91	59.40±0.27
	Chem.	51.30±1.29	52.13±0.64	52.62±0.52	53.66±0.97	59.23±0.32
	Phy.	50.20±1.46	52.34±0.67	50.15±0.59	51.41±0.86	61.55±0.58
	Math.	50.89±1.34	52.47±0.71	53.22±0.44	53.83±0.87	60.33±0.61
ColBert	Bio.	49.23±1.92	52.52±1.24	43.49±0.51	44.99±1.50	59.76±1.50
	Geo.	49.40±1.63	52.39±1.05	41.88±0.94	43.57±1.65	58.91±1.42
	Chem.	49.10±1.45	52.15±1.12	42.41±0.92	44.15±1.49	58.73±1.35
	Phy.	48.11±1.76	52.28±1.09	40.55±0.87	42.28±1.62	61.12±1.23
	Math.	48.80±1.81	52.36±1.18	43.12±0.83	44.09±1.63	59.81±1.38
MXBAI	Bio.	49.18±2.16	53.41±0.27	55.75±0.87	56.04±1.19	59.77±1.65
	Geo.	49.55±1.88	53.22±0.40	53.01±0.66	53.83±1.24	58.89±1.53
	Chem.	49.62±1.42	52.95±0.47	54.12±0.74	54.49±1.13	58.70±1.40
	Phy.	48.58±1.36	53.20±0.54	52.46±0.68	52.76±1.22	61.33±1.45
	Math.	49.35±1.63	53.33±0.51	55.12±0.59	54.88±1.20	59.89±1.52
T5	Bio.	51.47±2.23	53.30±1.08	51.80±0.62	52.73±1.31	59.95±0.25
	Geo.	51.91±1.89	53.01±0.92	49.50±0.55	50.62±1.24	59.20±0.33
	Chem.	51.86±1.74	52.60±0.89	50.36±0.47	51.31±1.25	59.10±0.41
	Phy.	50.78±1.60	52.96±1.02	48.42±0.51	49.66±1.31	61.48±0.35
	Math.	51.55±1.95	53.13±1.00	51.02±0.46	51.40±1.29	60.01±0.39

A Counterfactual Synthetic (Question-Context-Answer) Sample.

Question: What are two viable large-scale energy storage technologies?
Golden Answer: Underground Hydrogen Storage (UHS) and Compressed Air Energy storage (CAES).
Context:

- *Passage 1*: Underground hydrogen storage (UHS) and compressed air energy storage (CAES) are two viable large-scale energy storage technologies for mitigating the intermittency of wind and solar power. Therefore, it is meaningful to compare the properties of hydrogen and air with typical thermodynamic storage processes. This study employs a multi-physical coupling model to compare the operations of CAES and UHS, integrating gas thermodynamics within caverns, thermal conduction, and mechanical deformation around rock caverns. Gas thermodynamic responses are validated using additional simulations and the field test data. Temperature and pressure variations of air and hydrogen within rock caverns exhibit similarities under both adiabatic and diabatic simulation modes. Hydrogen reaches higher temperature and pressure following gas charging stage compared to air, and the ideal gas assumption may lead to overestimation of gas temperature and pressure. Unlike steel lining of CAES, the sealing layer (fibre-reinforced plastic FRP) in UHS is prone to deformation but can effectively mitigates stress in the sealing layer. In CAES, the first principal stress on the surface of the sealing layer and concrete lining is tensile stress, whereas UHS exhibits compressive stress in the same areas. Our present research can provide references for the selection of energy storage methods.
- *Passage 2*: Intensive increases in electrical energy storage are being driven by electric vehicles (EVs), smart grids, intermittent renewable energy, and decarbonization of the energy economy. Advanced lithium–sulfur batteries (LSBs) are among the most promising candidates, especially for EVs and grid-scale energy storage applications. In this topical review, the recent progress and perspectives of practical LSBs are reviewed and discussed; the challenges and solutions for these LSBs are analyzed and proposed for future practical and large-scale energy storage applications. Major challenges for the shuttle effect, reaction kinetics, and anodes are specifically addressed, and solutions are provided on the basis of recent progress in electrodes, electrolytes, binders, interlayers, conductivity, electrocatalysis, artificial SEI layers, etc. The characterization strategies (including in situ ones) and practical parameters (e.g., cost-effectiveness, battery management/modeling, environmental adaptability) are assessed for crucial automotive/stationary large-scale energy storage applications (i.e., EVs and grid energy storage). This topical review will give insights into the future development of promising Li–S batteries toward practical applications, including EVs and grid storage.
- *Passage 3*: To support increasing renewable capacity for a net-zero future, energy storage will play a key role in maintaining grid stability. In this paper, all current and near-future energy storage technologies are compared for three different scenarios: (1) fixed electricity buy-in price, (2) market-based electricity buy-in price, and (3) energy storage integrated into a fully renewable electricity system. In the first part of this study, an algorithm is devised to simulate strategic buy-in of electricity for energy storage. This analysis yields a qualitative decision-making tool for a given energy storage duration and size. Building upon the first part's findings, an integration study gives insight into expected power prices and expected storage size in a typical northwestern European fully renewable energy system. The integration study shows significant need for electricity storage with durations spanning from one to several days, typically around 40 h. Pumped Hydro Storage and Pumped Thermal storage surface as the best options. The overall levelized costs of storage are expected to be in the USD 200–500/MWh range. Integration of storage with renewables can yield a system-levelized cost of electricity of about USD 150/MWh. Allowing flexibility in demand may lower the overall system-levelized cost of electricity to USD 100/MWh."."Climate change mitigation requires the large-scale deployment of carbon capture and storage (CCS). Recent plans indicate an eight-fold increase in CCS capacity by 2030, yet the feasibility of CCS expansion is debated. Using historical growth of CCS and other policy-driven technologies, we show that if plans double between 2023 and 2025 and their failure rates decrease by half, CCS could reach 0.37 GtCO
- *Passage 4*: With the escalating utilization of intermittent renewable energy sources, demand for durable and powerful energy storage systems has increased to secure stable electricity supply. Redox flow batteries (RFBs) have received ever-increasing attention as promising energy storage technologies for grid applications. However, their broad market penetration is still obstructed by many challenges, such as high capital cost and inferior long-term stability. In this work, combining the merits of both all-vanadium and iron-chromium RFB systems, a vanadium-chromium RFB (V/Cr RFB) is designed and fabricated. This proposed system possesses a high theoretical voltage of 1.41 V while achieving cost effectiveness by using cheap chromium as one of the reactive species. Experimentally, the system attains a peak power density of over 900 mW cm-2 at 50°C and demonstrates stable performance for 50 cycles with an energy efficiency of over 87%, presenting this system as a promising candidate for large-scale energy storage.
- …
- *Passage 99*: Two large-scale energy storage technologies with limited safety: Sodium-ion batteries, which are known for their safety, and lithium-ion batteries, which have a higher risk of thermal runaway, are both unsuitable for use in a safe energy storage system.
- *Passage 100*: Two large-scale energy storage technologies with limited environmental sustainability: Lead-acid batteries, which are known for their environmental sustainability, and lithium-ion batteries, which have a higher environmental impact due to their production and disposal, are both unsuitable for use in an environmentally sustainable energy storage system.

Figure 9: A Counterfactual synthetic (Question, Context, Answer) Sample

A SSLI Synthetic (Question-Context-Answer) Sample.

Question: What is the most lethal melanoma subtype?
Golden Answer: uveal melanomas
Context:

- *Passage 1*: Activating mutations in GNAQ/GNA11 occur in over 90% of uveal melanomas (UMs), the most lethal melanoma subtype; however, targeting these oncogenes has proven challenging and inhibiting their downstream effectors show limited clinical efficacy. Here, we performed genome-scale CRISPR screens along with computational analyses of cancer dependency and gene expression datasets to identify the inositol-metabolizing phosphatase INPP5A as a selective dependency in GNAQ/11-mutant UM cells in vitro and in vivo. Mutant cells intrinsically produce high levels of the second messenger inositol 1,4,5 trisphosphate (IP3) that accumulate upon suppression of INPP5A, resulting in hyperactivation of IP3-receptor signaling, increased cytosolic calcium and p53-dependent apoptosis. Finally, we show that GNAQ/11-mutant UM cells and patients' tumors exhibit elevated levels of IP4, a biomarker of enhanced IP3 production; these high levels are abolished by GNAQ/11 inhibition and correlate with sensitivity to INPP5A depletion. Our findings uncover INPP5A as a synthetic lethal vulnerability and a potential therapeutic target for GNAQ/11-mutant-driven cancers.
- *Passage 2*: Background Although anti-Program-Death-1 (PD-1) is a standard adjuvant therapy for patients with resected melanoma. We hypothesize that there are discrepancies of survival, recurrence pattern and toxicity to adjuvant PD-1 between different ethnicities and melanoma subtypes. Objective/methodsWe performed a multicenter cohort study incorporating 6 independent institutions in Australia, China, Japan, and US.The primary outcomes were RFS and OS.Secondary outcomes were disease recurrence patterns and toxicities. ResultsIn total 534 patients were included.East-Asian/Hispanic/African had significantly poorer RFS/OS.Non-acral-cutaneous/unknown primary (NAC/UP) had the best RFS/OS, followed by acral; mucosal the poorest.Within the NAC/UP subtypes, East-Asian/Hispanic/African had significantly poorer RFS/OS than Caucasian.In the multivariate analysis incorporating ethnicity/melanoma-subtype/age/sex/stage/LDH/BRAFmutation/adjuvant-radiotherapy, East-Asian/Hispanic/African had independently significantly poorer outcomes (RFS: HR1.71 95%CI 1.19-2.44;OS: HR2.34, 95%CI 1.39-3.95),as was mucosal subtype (RFS: HR3.25, 95%CI 2.04-5.17;OS: HR3.20, 95%CI 1.68-6.08).Mucosal melanoma was an independent risk factor for distant-metastasis, especially liver metastasis.East-Asian/Hispanic/African had significantly lower incidence of GI/musculoskeletal/respiratory/other-rare-type-toxicities; but higher incidences of liver toxicities. Limitations retrospective study.Conclusions Ethnicity and melanoma subtype are associated with survival and recurrence pattern in melanoma patients treated with adjuvant anti-PD-1.Toxicity profile differs by ethnicity, and may require a precision toxicity surveillance strategy.
- *Passage 3*: Melanoma is the third most common type of skin cancer, characterized by its heterogeneity and propensity to metastasize to distant organs. Melanoma is a heterogeneous tumor, composed of genetically divergent subpopulations, including a small fraction of melanoma-initiating cancer stem-like cells (CSCs) and many non-cancer stem cells (non-CSCs). CSCs are characterized by their unique surface proteins associated with aberrant signaling pathways with a causal or consequential relationship with tumor progression, drug resistance, and recurrence. Melanomas also harbor significant alterations in functional genes (BRAF, CDKN2A, NRAS, TP53, and NF1). Of these, the most common are the BRAF and NRAS oncogenes, with 50% of melanomas demonstrating the BRAF mutation (BRAF
- *Passage 4*: Melanoma, the deadliest form of skin cancer, poses a significant clinical challenge for the development of effective treatments. Conventional in vivo animal studies have shown limited translational relevance to humans, raising strength to pre-clinical models for melanoma research. This review provides an in-depth analysis of alternative pre-clinical models including in vitro and ex vivo platforms such as reconstructed skin, spheroids, organoids, organotypic models, skin-on-a-chip, and bioprinting. Through a comprehensive analysis, the specific attributes, advantages, and limitations of each model are elucidated. It discusses the points related to the uniqueness advantages, from capturing complex interactions between melanoma cells and their microenvironment to enabling high-throughput drug screening and personalized medicine approaches. This review is structured covering firstly the roadmap to identify the co-occurrence of discovering new melanoma treatments and the development of its models, secondly it covers a comparative between the most used models followed by a section discussing each of them: the in vitro and ex vivo models. It intends to serve as an asset for researchers of melanoma field and clinicians involved in melanoma therapy, offering insights into the diverse preclinical models available for optimizing their integration into the translational pipeline.
- *Passage 5*: Melanoma incidence and mortality rates are historically higher for men than women. Although emerging studies have highlighted tumorigenic roles for the male sex hormone androgen and its receptor (AR) in melanoma, cellular and molecular mechanisms underlying these sex-associated discrepancies are poorly defined. Here, we delineate a previously undisclosed mechanism by which androgen-activated AR transcriptionally upregulates fucosyltransferase 4 (FUT4) expression, which drives melanoma invasiveness by interfering with adherens junctions (AJs). Global phosphoproteomic and fucoproteomic profiling, coupled with in vitro and in vivo functional validation, further reveal that AR-induced FUT4 fucosylates L1 cell adhesion molecule (L1CAM), which is required for FUT4-increased metastatic capacity. Tumor microarray and gene expression analyses demonstrate that AR-FUT4-L1CAM-AJs signaling correlates with pathological staging in melanoma patients. By delineating key androgen-triggered signaling that enhances metastatic aggressiveness, our findings help explain sex-associated clinical outcome disparities and highlight AR/FUT4 and its effectors as potential prognostic biomarkers and therapeutic targets in melanoma.
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- *Passage 99*: Deceptive: The most lethal melanoma subtype is often misdiagnosed as a harmless freckle, making early detection challenging.
- *Passage 100*: Perplexing: It's a common misconception that the most lethal melanoma subtype is the one that appears as a mole.

Figure 10: A SSLI synthetic (Question, Context, Answer) Sample

A Multi-Hop Synthetic (Question-Context-Answer) Sample.

Question: What are the isoforms categories of genes that control activity of gene transcription?
Golden Answer: rewirers and negative regulators
Context:

- *Passage 1*: Most human Transcription factors (TFs) genes encode multiple protein isoforms differing in DNA binding domains, effector domains, or other protein regions. The global extent to which this results in functional differences between isoforms remains unknown. Here, we systematically compared 693 isoforms of 246 TF genes, assessing DNA binding, protein binding, transcriptional activation, subcellular localization, and condensate formation. Relative to reference isoforms, two-thirds of alternative TF isoforms exhibit differences in one or more molecular activities, which often could not be predicted from sequence. We observed two primary categories of alternative TF isoforms: ŕewirersänd n̈egative regulators, both of which were associated with differentiation and cancer. Our results support a model wherein the relative expression levels of, and interactions involving, TF isoforms add an understudied layer of complexity to gene regulatory networks, demonstrating the importance of isoform-aware characterization of TF functions and providing a rich resource for further studies.
- *Passage 2*: More than 1600 human transcription factors orchestrate the transcriptional machinery to control gene expression and cell fate. Their function is conveyed through intrinsically disordered regions (IDRs) containing activation or repression domains but lacking quantitative structural ensemble models prevents their mechanistic decoding. Here we integrate single-molecule FRET and NMR spectroscopy with molecular simulations showing that DNA binding can lead to complex changes in the IDR ensemble and accessibility. The C-terminal IDR of pioneer factor Sox2 is highly disordered but its conformational dynamics are guided by weak and dynamic charge interactions with the folded DNA binding domain. Both DNA and nucleosome binding induce major rearrangements in the IDR ensemble without affecting DNA binding affinity. Remarkably, interdomain interactions are redistributed in complex with DNA leading to variable exposure of two activation domains critical for transcription. Charged intramolecular interactions allowing for dynamic redistributions may be common in transcription factors and necessary for sensitive tuning of structural ensembles.
- *Passage 3*: Transcription factors (TFs) control specificity and activity of gene transcription, but whether a relationship between these two features exists is unclear. Here we provide evidence for an evolutionary trade-off between the activity and specificity in human TFs encoded as submaximal dispersion of aromatic residues in their intrinsically disordered protein regions. We identified approximately 500 human TFs that encode short periodic blocks of aromatic residues in their intrinsically disordered regions, resembling imperfect prion-like sequences. Mutation of periodic aromatic residues reduced transcriptional activity, whereas increasing the aromatic dispersion of multiple human TFs enhanced transcriptional activity and reprogramming efficiency, promoted liquid–liquid phase separation in vitro and more promiscuous DNA binding in cells. Together with recent work on enhancer elements, these results suggest an important evolutionary role of suboptimal features in transcriptional control. We propose that rational engineering of amino acid features that alter phase separation may be a strategy to optimize TF-dependent processes, including cellular reprogramming.
- *Passage 4*: Determining whether the RNA isoforms from medically relevant genes have distinct functions could facilitate direct targeting of RNA isoforms for disease treatment. Here, as a step toward this goal for neurological diseases, we sequenced 12 postmortem, aged human frontal cortices (6 Alzheimer disease cases and 6 controls; 50% female) using one Oxford Nanopore PromethION flow cell per sample. We identified 1,917 medically relevant genes expressing multiple isoforms in the frontal cortex where 1,018 had multiple isoforms with different protein-coding sequences. Of these 1,018 genes, 57 are implicated in brain-related diseases including major depression, schizophrenia, Parkinson's disease and Alzheimer disease. Our study also uncovered 53 new RNA isoforms in medically relevant genes, including several where the new isoform was one of the most highly expressed for that gene. We also reported on five mitochondrially encoded, spliced RNA isoforms. We found 99 differentially expressed RNA isoforms between cases with Alzheimer disease and controls.
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- *Passage 99*: Pioneer transcription factors (TFs) exhibit a specialized ability to bind to and open closed chromatin, facilitating engagement by other regulatory factors involved in gene activation or repression. Chemical probes are lacking for pioneer TFs, which has hindered their mechanistic investigation in cells. Here, we report the chemical proteomic discovery of electrophilic small molecules that stereoselectively and site-specifically bind the pioneer TF, FOXA1, at a cysteine (C258) within the forkhead DNA-binding domain. We show that these covalent ligands react with FOXA1 in a DNA-dependent manner and rapidly remodel its pioneer activity in prostate cancer cells reflected in redistribution of FOXA1 binding across the genome and directionally correlated changes in chromatin accessibility. Motif analysis supports a mechanism where the covalent ligands relax the canonical DNA binding preference of FOXA1 by strengthening interactions with suboptimal ancillary sequences in predicted proximity to C258. Our findings reveal a striking plasticity underpinning the pioneering function of FOXA1 that can be controlled by small molecules.
- *Passage 100*: Alternative transcription start site usage (ATSS) is a widespread regulatory strategy that enables genes to choose between multiple genomic loci for initiating transcription. This mechanism is tightly controlled during development and is often altered in disease states. In this review, we examine the growing evidence highlighting a role for transcription start sites (TSSs) in the regulation of mRNA isoform selection during and after transcription. We discuss how the choice of transcription initiation sites influences RNA processing and the importance of this crosstalk for cell identity and organism function. We also speculate on possible mechanisms underlying the integration of transcriptional and post-transcriptional processes.

Figure 11: A multi-hop synthetic (Question, Context, Answer) sample.